Development of High Temperature High Strength Aluminium Alloys

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Abstract

The combined use of cerium and scandium in a cast Al-Ce-Sc alloy for use at high temperatures is investigated using the ThermoCalc computational software. Three alloys are tested for both strength and thermal stability - Al-10%Ce-0.2%Sc,0.5%Sc and 1%Sc. All present a microstructure composed of eutectic $Al_{11}Ce_3$ and α-Al phases with the latter being reinforced by L12 Al_3Sc intermetallic precipitates. It is found that, in all cases, interfacial diffusion of alloying elements between the phases is negligible. The Al_3Sc precipitates are remarkably stable at temperatures reaching 300°C being able to maintain up to 96% of their strength after two days. Yet, their coarsening is gravely accelerated at higher temperatures with reasonable service life of a few minutes at 350°C. Promising results on the viability of the alloying system are obtained yet further work is required to validate its potential as a high strength, high temperature aluminium alloy.

Keywords: Aluminium Alloy, High Temperature, Precipitation Strengthening, Eutectic, Thermal Stability, Scandium, Cerium

Introduction

Since their introduction into mainstream use at the turn of the XXth century, aluminium alloys have found a large number of applications in fields ranging from soda cans to aerospace thanks to their relatively low weight and acceptable mechanical properties. Nevertheless, their use at higher temperatures is limited by their poor thermal stability. While aluminium's fusion point of 660°C intrinsically limits its use above temperatures of 400°C, most high strength aluminium alloys suffer from poor thermal stability at temperatures exceeding the 150 to 200°C range. As an example, the wrought 7xxx Al-Zn-Mg-Cu alloys reputed for their high strength cannot be used at temperatures above 150°C due to Ostwald ripening of their precipitates. In other example, the widely used Al-Si cast alloys of the 3xx.x and 4xx.x series similarly lose their mechanical resistance at similar temperatures. Aluminium alloys have thus been replaced by more expensive yet similarly corrosion resistant titanium alloys in high strength applications.

Yet, the higher costs of titanium alloys have motivated recent research into the possibility of manufacturing high strength and thermally stable aluminium alloys. The purpose of the presented work is to explore and propose a possible aluminium alloy for high strength applications.

Review of Thermally Stable Al-Alloys

Wrought Al-Alloys

The majority of high strength wrought aluminium alloys suffer from similar weakening at high temperature. Wrought aluminium alloys can be categorised into two families - precipitation hardened and non precipitation hardened alloys. The former, which possess the better mechanical properties, suffer from severe strength degradation at temperatures above 150°C. This can be explained by the coarsening of the precipitates whose initial optimal size is lost as thermally activated diffusion of alloying elements and Ostwald ripening occur [1]. For instance, the 6061 aluminium alloy composed of aluminium and Mg_2Si precipitates sees its yield strength decrease from 325 MPa at room temperature to 275 at 200°C before suffering from severely decreasing down to 175 MPa at 250°C. Similar behaviour is observed for the 7075 aluminium alloy ubiquitous in the aerospace industry [2]. Non precipitation hardened alloys also suffer from similar weakening at high temperatures due to to temperature induced grain coarsening. For instance, the 5083 alloy with magnesium as a primary solid solution strengthener suffers from a similar decrease in yield strength, going from 275 MPa at room temperature to 150 MPa at 250°C [3].

Most current research endeavours are aimed at two different possible solutions. The first regards the strengthening of current aluminium alloys at high temperatures of which a notable example is the creation of the Al-Cu-Mg-Ag alloy which maintains its mechanical properties at temperatures above 150°C [4].

The second axis of research revolves around the elaboration of new primary alloying elements to replace the current elements used for precipitate formation. All precipitates considered for this role are L12 ordered intermetallic. They are preferable due to their coherent relationship with the FCC aluminium matrix and their proven effectiveness as high-temperature strengthening phases, analogous in structure to the γ' precipitates used in nickel-based superalloys [2]. The alloying element must also have low diffusion coefficients into the aluminium matrix. The main alloying elements being considered with this regard are zir-

conium, scandium and vanadium. With regards to the latter, the possibility of forming Al_3V precipitates does exist. Yet, vanadium cannot be reliably utilised as a primary alloying element in spite of its low diffusion coefficient on the order of $10^{-21}\ m^2.s^{-1}$ [2]. When large amounts of vanadium are used as alloying elements, it has a tendency to either form large $Al_{10}V$ precipitates with a non L12 structure or to dramatically coarsen creating weak points in the matrix and reducing its strength at high temperatures [5]. Furthermore, vanadium precipitates have poor coherency with the aluminium matrix [6], limiting its usefulness as an alloying element in spite of the benefits microadditions of vanadium may have.

Such issues are not present in either zirconium or scandium with both materials forming coherent L12 precipitates in aluminium [6] and thus acting as the most promising alloying elements for the formation of L12 precipitates in high temperature aluminium alloys [2]. Both materials have very low diffusion coefficients in aluminium on the order of $10^{-15}\ m^2.s^{-1}$ for scandium [2] and $10^{-21}\ m^2.s^{-1}$ for zirconium [7]. While both are frequently used in conjunction to one another, they each come with their own benefits and disadvantages. Zirconium, while much cheaper than scandium, necessitates very long heat treatments for precipitation formation due to its low diffusion coefficient [8]. Furthermore, the Al_3Zr L12 precipitate is only thermodynamically metastable at room and high temperatures. While not causing any issues with regards to its production and use, this property does significantly reduce the feasibility of studying it using the methodology described in part 3 of this report. These issues do not however impact scandium whose reasonable heat treatment duration and coherency in the matrix [6] [9]. Alloying scandium with aluminium further endows the alloy with excellent creep resistance greater than that of pure commercial alloy and the majority of aluminium alloys typical in industry [10]. These properties of scandium make it an ideal candidate for a potential high temperature and high strength aluminium alloy.

Cast Al-Alloys

Thanks to its easily accessible eutectic point at 12.6wt.% Si and low cost, the Al-Si system remains the most widely utilised cast aluminium alloy to date. Its thermal stability at high temperature is however limited, first by its low melting temperature of 570-600°C. Traditional Al-Si alloys present a microstructure composed of coarse brittle needle like silicon dispersed in an aluminium matrix providing some strengthening at room temperature [11]. At temperatures above 150°C, silicon's relatively high diffusion coefficient on the order of $10^{-13}\ m^2.s^{-1}$ in aluminium leads to both a coarsening and spherodisation of already present eutectic [11] [12] and precipitation of the silicon dissolved in the aluminium matrix leading to severe decreases in strength. Other aluminium alloys containing silicon such as the A380 Al-Cu-Si hypo-eutectic alloy is similarly brittle at high temperatures due to coarsening of the Al_2Cu precipitates [13].

Current solutions aimed at remedying these issues are centred in two main axes.

The first is the reinforcement of the traditional Al-Si system with transition metals such as iron, nickel, zinc or silver in order to form temperature stable L12 Al_3X precipitates in the aluminium matrix [2] as discussed for wrought aluminium alloys. Vanadium for instance has proven itself to provide highly stable $Al_{10}V$ precipitates that enable AlSiV alloys to maintain their mechanical properties at 200°C [14]. The same goes for titanium and zirconium, with microadditions of these three alloying elements extending the thermal stability of traditional cast Al-Si alloys up to temperatures of 200°C [15].

The second axis revolves around the complete replacement of silicon in the Al-Si system [2]. The requirements for such a replacement material are the presence of an easily accessible eutectic for strengthening and that the element be cheap and easily accessible. Titanium and nickel both provide possible alternatives to silicon. Nickel in particular provides an easily accessible eutectic of 10-12wt.% at a similar temperature of 640°C but with greatly reduced diffusion on the order of $10^{-14}\ m^2.s^{-1}$ in aluminium, endowing Al-Ni alloys with great thermal stability [16].

An alternative to nickel lay in cerium [2] [17] [18], a cheap rare earth element of the lanthanide series with an easily accessible eutectic at 10.6wt.% Ce. Furthermore, the large size mismatch between cerium and aluminium leads to the former having low solubility in aluminium and a low diffusion coefficient on the order of $10^{-18}\ m^2.s^{-1}$. Aluminium cerium alloys present a microstructure composed of an α-Al matrix coupled with an $Al_{11}Ce_3$ eutectic matrix. The latter is much harder than aluminium thus endowing raw Al-Ce alloys with highly inhomogeneous mechanical properties similar to that of traditional composites [18] [19] that do however remain very stable at high temperature. The addition of a combination yttrium, zirconium or magnesium have however been proven to be promising at reducing the strength mismatch in phases through solid solution strengthening thereby making Al-Ce systems more mechanically stable [19]. Proper reinforcement of the α-Al matrix in Al-Ce alloys coupled with the eutectic's high thermal stability at temperatures reaching 500°C [2] [20] would thus lead to the formation of a highly temperature stable and high strength aluminium alloy.

System Choice

The literature review described in the previous section highlights two cerium and scandium as two promising alloying elements for future high strength and high thermal stability aluminium alloys. Yet, they both possess their own downsides with the use of cerium as an alloying element leading to the formation of a eutectic microstructure with severe strength mismatches and scandium, in spite of its good thermal stability and strengthening, being prohibitively expensive.

The natural solution that follows would thus be to combine the two alloying elements to create a cast Al-Ce-Sc alloy with the goal of combining the high strength $Al_{11}Ce_3$ eutectic with that of α+Al_3Sc to remedy the weakness of standard Al-Ce alloys. Alloys with similar yet more complex design and composition have already been studied in literature [21] [22]. The goal of the project work is thus to ensure the strength and high temperature stability of the proposed alloy.

Methodology

Computational Method

All computational studies presented below are performed using the ThermoCalc software suite [23].

An initial study of the ternary Al-Ce-Sc system is performed in order to obtain the value of the eutectic composition of the alloy and to ensure the presence and thermodynamic stability of the expected phases. The latter are ensured to occur at the weight fractions of both alloying elements in literature corresponding to around 10%wt. for cerium and around 1%wt. for scandium.

Further studies of scandium and cerium must be performed in separate workflows due to their intrinsic differences as alloying elements in aluminium. Being a precipitation alloying element, scandium is studied using Thermocalc's precipitation module. This allows for a preliminary determination of the rough heat treatment that must be performed in order to obtain optimally sized Al_3Sc precipitates in the α-Al matrix. The thermal stability of these precipitates is evaluated using the diffusion module.

The strength of the $Al_{11}Ce_3$ eutectic phase cannot however be studied using ThermoCalc. Literature values will thus be used instead. The stability of the $Al_{11}Ce_3$ phase will however be analysed in the same manner as that of the Al_3Sc precipitates with the same expectations for high temperature stability.

Alloy Design Criteria and Objectives

The temperature ranges aimed for are described in the following paragraph yet depend on both literature and an initial analysis of the phases' stability at the sought after compositions and temperature. Nevertheless, the following criteria will be used to evaluate the strength and thermal stability of the alloying system.

- Interphase diffusion of alloying elements is low to negligible over long periods of time at service temperature
- In the case of Al_3Sc precipitates, coarsening must lead to acceptable decreases in strength of 10% or less at service temperature over a period of two days
- The strength of the Al_3Sc precipitates matches that of $Al_{11}Ce_3$ thereby remedying the strength mismatches

Results

Phase Stability and System Choice

Figure (1) indicates that all of the studied systems demonstrate thermodynamic stability of all expected phases at the compositions and temperatures investigated with ThermoCalc correctly predicting the presence of α-Al, Al_3Sc and $Al_{11}Ce_3$ [2].

The eutectic composition of cerium in the alloy remains constant at 10%wt. cerium over all of the tested compositions. This value is consistent with literature [18].

The tools at hand do not permit an in depth analysis of the microstructure of the α-Al/$Al_{11}Ce_3$ eutectic system. As such, the cerium content of the alloy is fixed at the eutectic point of 10%.wt throughout all simulations with the scandium content of

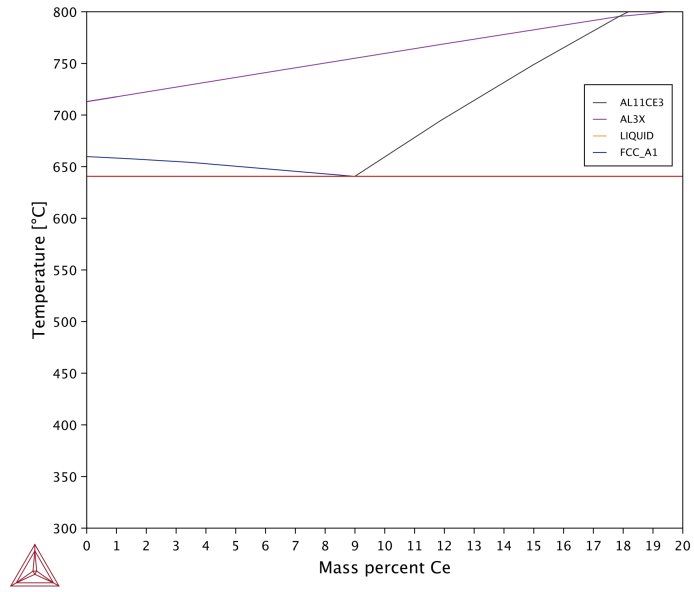


Figure 1: Example phase diagram with 1%.wt Sc fixed. It should be noted that no intermetallic phases are found to be composed of both scandium and cerium.

the alloy being allowed to vary with compositions of 0.2%.wt , 0.2%.wt and 1%.wt Sc being chosen based off of values observed in literature [2] [8] [9] [21].

Figure (1) returns a melting temperature of 660°C for the alloy. Considering that current high strength aluminium alloys may be used at temperatures of up to 200°C and literature, a target temperature range of 300 to 350°C is chosen for the new alloys with all simulations being conducted at both temperatures.

Of note is the small solubility of cerium in aluminium which can immediately be noticed through the absence of any pure α domain in figure (1). Theoretical modelling returns that cerium has a high temperature solubility of at most 10^{-4}%.wt[1] in aluminium as a result of cerium's large size mismatch and low diffusion coefficient in aluminium.'

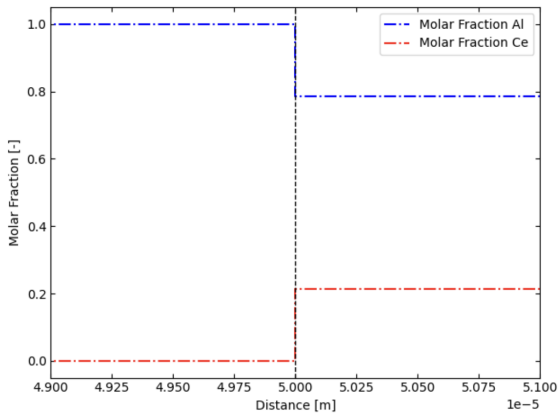
Stability of $Al_{11}Ce_3$

$Al_{11}Ce_3$ forms a continuous separate phase to the α-Al. The eutectic is an intermetallic with high strength being documented in literature as going up to 350 HV [18]. The strength of a continuous eutectic being impossible to ascertain with the tools at hand, it is assumed on the basis of literature to have high strength and hardness at the sought after temperatures [2]

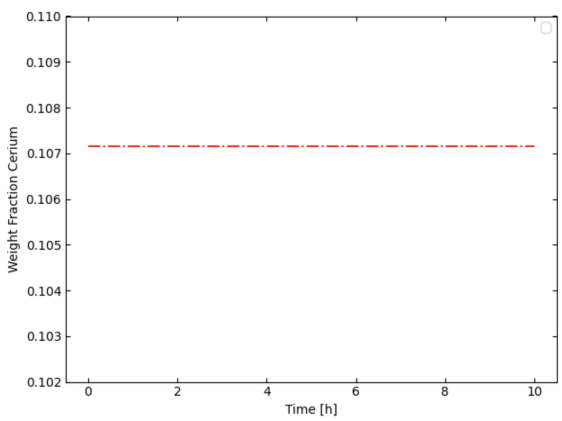
The intermetallic has a melting point of 1251°C and a reported low diffusion on the order of $10^{-18}\ m^2.s^{-1}$. It is hypothesised based on existing literature that the intermetallic remains stable in microstructure at the sought after temperature range of 300-350°C while maintaining its mechanical properties [2]. Figures (6) highlight the results of diffusion simulations of cerium in the Al-10%Ce system. The $Al_{11}Ce_3$ phase is shown to be exceptionally stable at 350°C. The interface between $Al_{11}Ce_3$ and α-Al shows little to no diffusion on a micrometre scale of either cerium

[1] This value was provided to us during communications with Dr. Nicholas Grundy from ThermoCalc

into α-Al or aluminium into $Al_{11}Ce_3$ over 2 days at 350°. Stability of the eutectic composition is further highlighted by the invariance over time of the weight fraction of cerium in the eutectic structure.



(a)



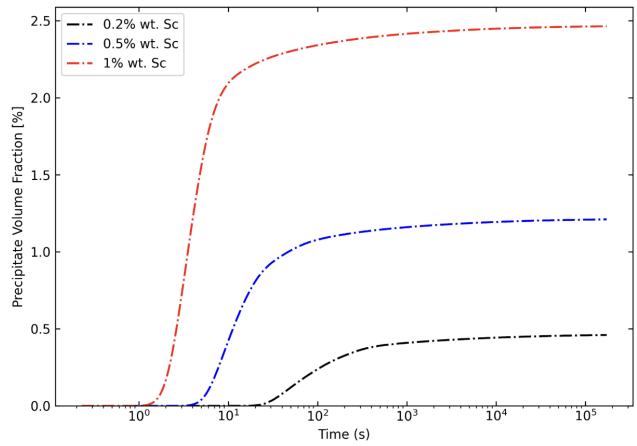
(b)

Figure 2: Thermal stability of cerium and $Al_{11}Ce_3$ over two days at 350°C. (a) highlights the weak and quasi-non existent diffusion between $Al_{11}Ce_3$ and α-Al and (b) shows the stability of the eutectic phase's composition

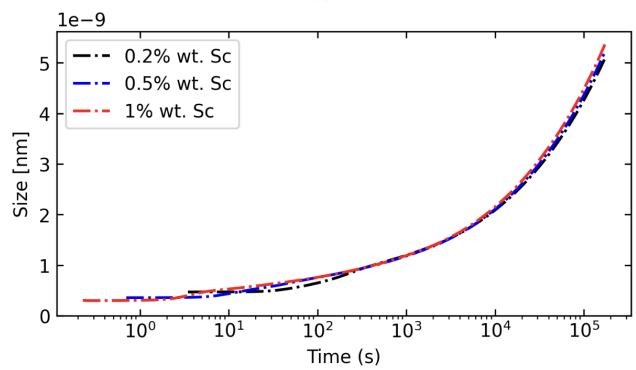
Strength and Stability of Al_3Sc

Scandium is used to form high strength and temperature stable L12 precipitates to strengthen $\alpha-$Al in front of the stronger and harder $Al_{11}Ce_3$ eutectic phase. As such, scandium is studied in the $\alpha-Al_3Sc$ system.

The predominant use of cerium as an alloying element necessitates the alloy production through cast instead of wrought means. Nevertheless, adequate heat treatments starting with an initial dissolution of the scandium into the aluminium matrix followed by heating cycles can be used to form scandium precipitates in aluminium. Al_3Sc being thermodynamically stable precipitates as shown in figure (1), heat treatments are not mandated to occur at higher temperatures than the service temperature. Furthermore, precipitation heat treatments usually being performed at 0.45-0.55T_m [1], ageing would occur at around the sought after temperatures. As a result, heat treatment for precipitation formation is designed to occur at a temperature of 350°C. Figures (7) demonstrate that varying the scandium content from 0.2 to 1%.wt leads to overall alloy yield strength ranging from 170 to 380 MPa with optimal precipitate sizes of 3-3.12 nm. Optimal precipitation strengthening is achieved after heat treatments of over 8 hours. Precipitate volume fractions are found to increase with the scandium content of the alloy peaking at 2.5% for 1%.wt Sc achieved after only a few seconds of heat treatment. At 0.2%.wt, precipitate volume fraction is of at most 0.5% achieved over a five minutes of heat treatment.



(a)



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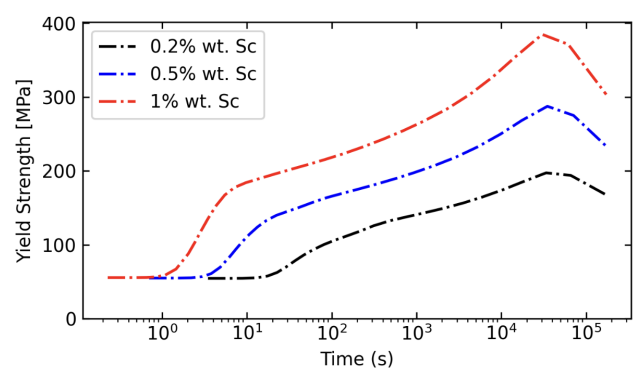

(c)

Figure 3: Heat treatment results for Al_3Sc formation at 350°C with (a) the evolution of precipitate volume fraction over time, (b) the size of the precipitates and (c) the resulting yield strength of $\alpha-$Al-Al_3Sc

Analysis of the precipitates' thermal stability is performed through the diffusion module as was done for cerium and by studying the evolution of the precipitates' size over time. The results of the former are shown in figure (4). Identical conclusions as to the diffusion of cerium into aluminium can be made with negligible scandium diffusion at the interface between precipitates and the matrix. The size of the precipitates is studied using the Lifshitz–Slyozov–Wagner (LSW) coarsening law in equation (1) below where r_0 is taken as the post heat treatment optimal size and K is the Ostwald ripening rate. The ripening rate is computed using data computed with ThermoCalc on the evolution of precipitate size over time at a certain temperature shown in the annex.

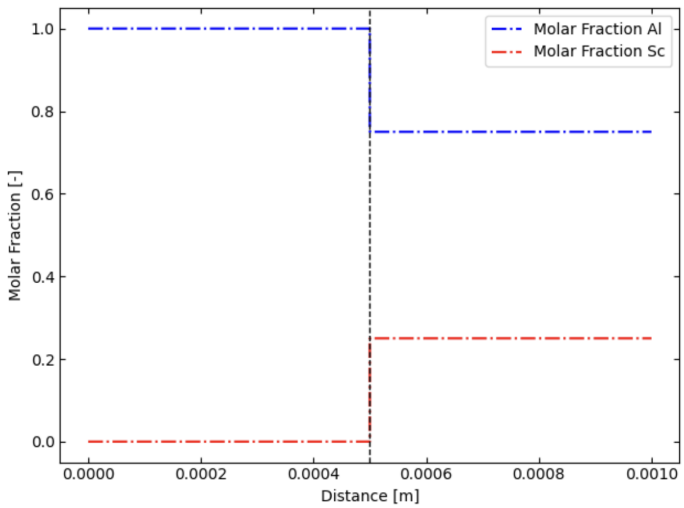

Figure 4: Diffusion of scandium from Al_3Sc into α-Al after 2 days at 350°C

$$r^3 = r_0^3 + K \cdot t \tag{1}$$

Ostwald ripening rates of an average of 0.026 and 0.087 nm.$s^{-1/3}$ are respectively found for the alloys at 300 and 350°C. Precipitate coarsening is thus estimated at 300 and 350°C in figures (5). It is immediately evident that coarsening is much more severe at 350°C with the precipitates nearly doubling in size in 2 days compared to the 50% increase observed at 300°C. The size data is then correlated with that shown in figures (7) to extract the evolution of α-Al_3Sc yield strength over time represented in table (1).The computations thus show that all α-Al_3Sc are able to maintain up to 97% of their room temperature yield strength for up to 2 days at 300°C. Similar behaviour is only observed on shorter time scales at 350°C however with all of the alloys suffering severe strength decreases of more than 10% in 3 hours and up to 40% over 2 days.

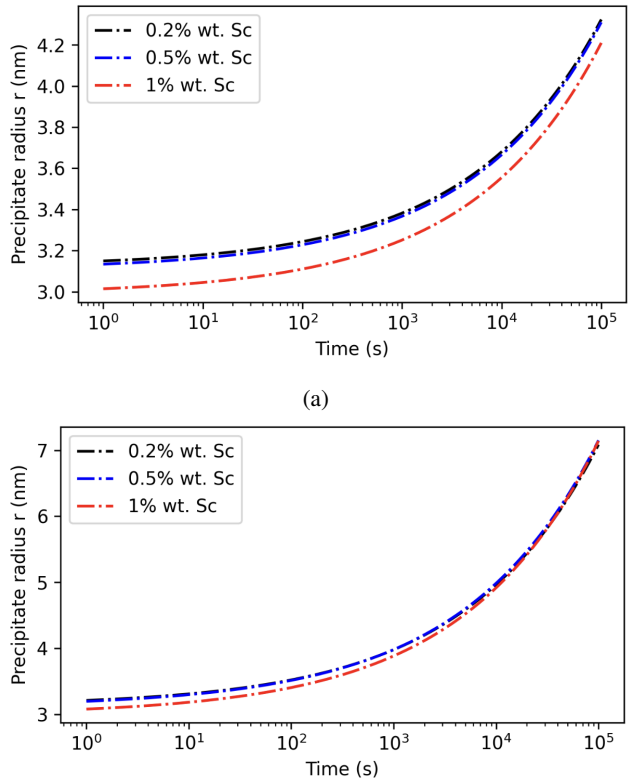

Figure 5: LSW computed coarsening of the Al_3Sc precipitates over time at (a) 300°C and (b) 350°C

Discussion

The study on the $Al_{11}Ce_3$ phase reveals that it has almost perfect thermal stability. Negligible diffusion is observed at the interface between the $Al_{11}Ce_3$ and $\alpha-$Al phases suggesting that the alloy will maintain its overall structure and stability at the high service temperatures at play. This result is in line with literature suggesting overall stability of the alloy at temperatures reaching and exceeding 300°C with $Al_{11}Ce_3$ being highly resistant to coarsening. It should however be noted that extensive literature suggests that the Al-Ce system will lose significant hardness when reaching temperatures above 400°C [2] [18] [20]. While not studied in the scope of this project, thermal dependence of $Al_{11}Ce_3$ could hamper the alloy's overall thermal stability although a more detailed study is required before any definitive conclusions can be made. Furthermore, while the strength of the alloy is known to increase with the cerium content [18], hyper-eutectic compositions lead to incoherent phase boundaries between the eutectic phases thus leading to weakened overall alloy strengthening in spite of $Al_{11}Ce_3$'s high strength. This effect is not studied in any more detail in the scope of this project.

Identical conclusions can be made with regards to scandium's thermal stability with negligible scandium diffusion being observed into the $\alpha-$Al matrix in line with its low diffusion coefficient. The severe coarsening observed in Al_3Sc precipitates at 350°C [24] is in accordance with literature on the topic and the computed Orowan coarsening constants of 0.026 and 0.087 nm.$s^{-1/3}$ are in full accordance with experimental values [25]. The study of precipitate coarsening presented in figures (5) and table (1) clearly demonstrates that the performed computational studies imply, in line with theory that the Al-Sc system at hand can be used at 300°C without long term strength degradation. It should however be noted that strength losses at 300°C grow more severe with decreasing scandium content at 8-14% for 0.2 and

Table 1: Relative yield strength degradation due to precipitate coarsening at 300 °C and 350 °C, referenced to the peak yield strength for each composition.

Composition	Model time [s]	α-Al_3Sc Yield strength [MPa]		Decrease from peak [%]	
		300 °C	350 °C	300 °C	350 °C
0.2 wt% Sc	10^3	166.9	159.7	0.7	5.0
	10^4	163.9	144.5	2.5	12.8
	10^5	153.9	128.1	8.4	23.8
0.5 wt% Sc	10^3	278.3	266.3	0.0	4.3
	10^4	266.3	237.9	4.3	14.5
	10^5	237.9	177.6	14.5	36.2
1.0 wt% Sc	10^3	378.5	365.8	0.0	3.3
	10^4	365.8	323.0	3.3	14.7
	10^5	365.8	228.3	3.3	39.7

0.5%.wt Sc. This could prove to be problematic with the high cost of scandium motivating its potential partial substitution with other alloying elements such as zirconium. The alloy system's service temperature remains capped at 350°C no matter the composition. Use at such temperature leads to severe coarsening that is heavily detrimental to the alloy's long term use with a nearly 15% reduction in yield strength after 2 hours at high temperature. Such behaviour is consistent with that observed for other precipitation strengthened aluminium alloys at higher temperatures such as the 6xxx and 7xxx systems [2] [26]. Furthermore, the predicted precipitate volume fractions remain below 2.5 vol% even for 1 wt% Sc additions. While modest in absolute terms, these values are consistent with reported Al–Sc alloys [25] and reflect the high strengthening efficiency of coherent L12 precipitates. The maximal computed yield stresses nearing 400 MPa reflect those attained by other, more heavily alloyed systems at room temperature such as the 7xxx series [2] or the A380 alloy [13]. The addition of scandium to form precipitates in aluminium alloys can thus be hypothesised to strengthen the $\alpha-$Al matrix enough to offset the strength imbalance between the two structures leading to the formation of a high strength and high temperature aluminium alloy [21] [22].

A final comment should be made on the proposed heat treatment for precipitate formation. It is, as previously mentionned, in accordance with precipitation heat treatments described in literature typically hovering in the 250-350°C range [9] [25] [27]. The albeit slightly higher than average temperature of 350°C enables rapid precipitate formation in 2 hours which should both decrease the size distribution of the precipitates and increase their uniformity in the α-Al matrix. An initial study involving a heat treatment of 450°C was conducted. Identical yield strengths and volume fractions are found to occur in a heat treatment lasting under two minutes. Basic metallurgical theory indicates that such a heat treatment would likely hamper the homogeneity of the precipitates and thus the alloy's overall strength. Experiments have also shown heat treatments at such temperatures lead to incoherent precipitates [27]. The computed coarsening is however found to be consistent with existing experimental results [24].

Conclusion

The studied alloy system Al-Ce-Sc is found to maintain its mechanical properties over long periods of time at temperatures of up to 300°C. This presents a significant improvement relative to current high temperature aluminium alloys with service temperatures capped at 200°C. The L12 Al_3Sc precipitates provide yield the α-Al_3Sc system with a total yield strength varying between 160 and 380 MPa with increasing scandium content corresponding to adequate precipitation strengthening of the matrix. Such strength should in principle remedy the severe strength mismatches between the $Al_{11}Ce_3$ and α phases traditionally observed in Al-Ce systems. This strength, coupled with the almost negligible diffusion of alloying elements even at high temperatures reaching 450°C, means that the Al-Ce-Sc system provides a promising platform for the creation of high strength high temperature aluminium alloys.

Further more detailed computational and experimental work is however required to obtain definitive heat treatments and compositions for future aluminium alloys. Some of this additional work could be focused more towards the Al-Ce system that, while being temperature stable, can lose some of its mechanical strength at elevated temperatures. The partial replacement of scandium by zirconium or erbium is also foreseeable as alloys with scandium proportions reaching 1%.wt would prove to be prohibitively costly for large scale applications in spite of offering the best mechanical and temperature resistance. Finally, the conducted work omitted all analysis of the corrosion resistance of the alloy.

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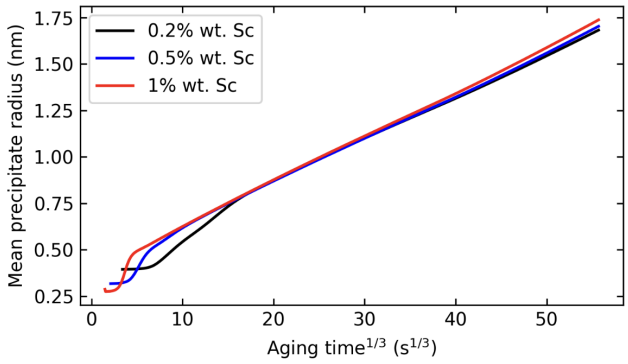
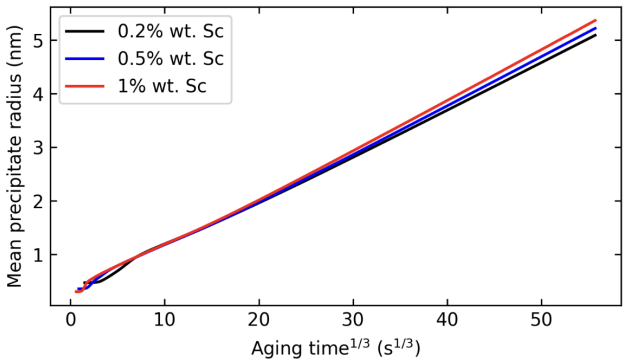
Appendix

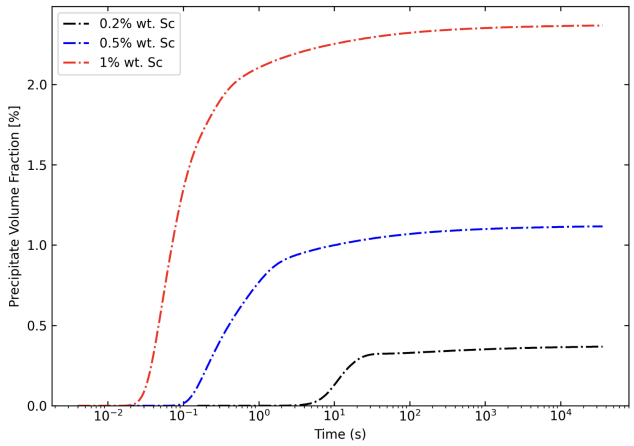


(a)

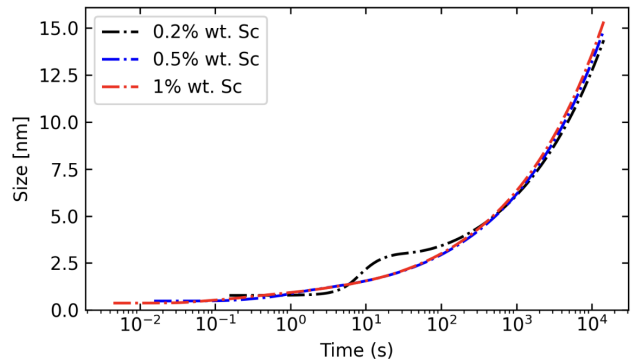


(b)

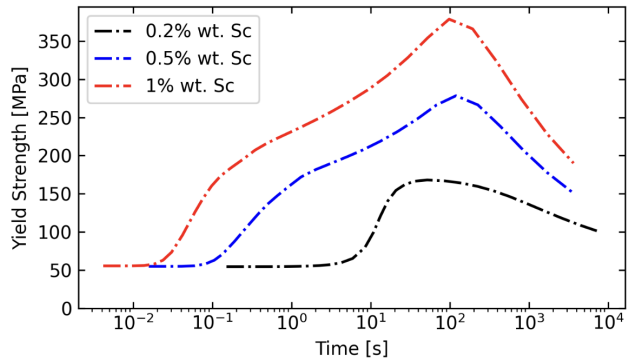
Figure 6: Linearisation of the size as a function of time curves obtained at (a) 300°C and (b) 350°C. The original curves as computed by ThermoCalc are identical in form to that shown in figure (5(b)). These are used to extract the computed Ostwald ripening coefficients and thus to obtain figures (5).



(a)



(b)



(c)

Figure 7: Heat treatment results for Al_3Sc formation at 450°C with (a) the evolution of precipitate volume fraction over time, (b) the size of the precipitates and (c) the resulting yield strength of $\alpha-$Al-Al_3Sc. It is immediately evident that optimal properties are reached after 100 seconds. Such a heat treatment would thus lead to an inhomogeneous microstructure with incoherent precipitates.